SYNTHETIC TEACHING DOCUMENT. Written for the STEMLink AI Engineer Bootcamp. Not agronomic or legal advice.

Paddy Cultivation Guide: Maha Season (Teaching Edition)

1. Choosing a variety

Varieties are grouped by age class, the number of months from sowing to harvest. Bg 300 is a three-month variety. Bg 352 is a three-and-a-half-month variety. Bg 403 is a four-month variety.

Three-month varieties suit fields where water may run short late in the season, because the crop is harvested before the tank level falls. Four-month varieties give the crop more time to fill grain but need water for longer.

Farmers should use certified seed paddy. Seed saved from a previous harvest often carries weed seed and has a lower germination rate.

2. Land preparation and sowing

Plough the field twice, with about two weeks between the first and second ploughing, so that weeds have time to germinate and be turned under. Level the field before sowing so that water stands at an even depth.

In broadcast fields, sow pre-germinated seed onto a puddled, drained surface. Keep the field moist but not flooded for the first ten days so the seedlings can root.

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3. Fertiliser timing

Apply the basal fertiliser on the day of sowing or the day before. Nitrogen is then split into top dressings so the crop receives it when it can use it.

For three-month varieties, apply the first top dressing two weeks after sowing and the second top dressing five weeks after sowing. For three-and-a-half and four-month varieties, apply the first top dressing two weeks after sowing and the second six weeks after sowing.

The quantity of each application must follow the approved fertiliser schedule for the district, issued through the Agrarian Service Centre. This guide does not give quantities.

Do not apply nitrogen when heavy rain is forecast within the next two days. Runoff over the field bunds carries the fertiliser away before the crop can take it up. Drain standing water and apply when the soil surface is moist but not flooded.

4. Yellowing and weak growth

Yellowing leaves and weak growth may be associated with nitrogen shortage, but the same signs appear with waterlogging, poor drainage, pests, disease and other nutrient problems. Yellowing alone does not confirm the cause.

When yellowing is seen, check whether it covers the whole field or only some sections, whether older or younger leaves are affected, whether water is standing in the field, and whether drainage channels are blocked.

5. Water management

Keep two to five centimetres of standing water from the start of tillering until two weeks before harvest, except when draining for a top dressing. Drain the field fully two weeks before harvest so that the grain ripens evenly and machines can enter.